MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, ROCKET ASSISTED, M549, INERT, SIMULATED PARTS AND LOADING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the fabrication of the inert parts and the complete inert round loading and assembly of Projectile M549.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-45607 - Inspection Equipment, supply and maintenance of.
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications for.

STANDARD

FEDERAL

FED-STD-151 - Metals, Test methods.

MILITARY

MIL-STD-1167 - Ammunition, Data Cards
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components
MIL-STD-453 - Inspection, Radiographic

DRAWINGS

ARMY

8837839 - Pallet for projectile
9236005 - 155MM, Inert, RAP assembly M549
9298010 - Grain Forward inert
9298011 - Grain, Aft, inert
9298012 - Housing, Delay, Dummy

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material and parts shall be in accordance with the applicable drawings and specifications.

3.2 The components, and assemblies shall comply with all requirements specified on drawing 9236005 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Loading of inert filler.

3.3.1 Mating surfaces.-The mating surfaces of threads shall be free of caked inert filler and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the filler and becoming entrapped in the warhead.

3.4 Assembly torque (Motor Body-Warhead).-The motor body shall be assembled to the warhead while the adhesive is still wet to meet the torque requirement specified on dwg.9236005. The equipment used for applying torque to the motor body-warhead assembly shall be calibrated at the start of each production shift as well as each time there is a change in equipment, settings or process that is likely to affect uniformity of performance as determined by the Government. The equipment shall be provided with a suitable indicator to aid in observance of proper torque application (see 4.4.3.1).

3.4.1 Disassembly torque (Lifting plug-Warhead)-If a lifting plug is assembled to the projectile in lieu of a fuze the assembly torque shall be sufficient to prevent loosening upon application of 100 in-lb minimum disassembly torque, and assure loosening upon application of 200 in-lb maximum disassembly torque.

3.5. Workmanship.-All manufacturing, processing and assembling operations including loading of the warhead shall be correctly performed in a thorough, workmanlike manner. Parts shall be free of burrs, chips, cracks, dirt, grease, rust, corrosion products and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent. Care shall be exercised at all times so that the rotating band of each motor body will not be damaged by being nicked or burred.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection and standard quality assurance provision.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

4.2 Classification of Inspection.-See MIL-A-48078.

4.2.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents. And that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3 First Article Inspection - See MIL-A-48078 (PA)

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Inert Grain Forward	9298010	15
Inert Grain Aft	9298011	15
Housing, Delay, Dummy	9298012	15

4.3.2 Inspections to be performed.-See MIL-A-48078 (PA)

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078 (PA).

4.4.2 Examinations.-See MIL-A-48078 (PA).

Sampling Plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

AGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Grain, Forward, Inert	1 OF 1	9298010
			NEXT HIGHER ASSEMBLY 9236005

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Outside Diameter		0.40%	3.2	Gage
132	Diameter of hole		0.40%	3.2	Gage
133	Depth of hole		0.40%	3.2	Gage
134	Overall length		0.40%	3.2	Gage
Minor					
201	Radius missing or improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298011
4.4.2.2	Grain, Aft, Inert				NEXT HIGHER ASSEMBLY 9236005
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Outside diameter		0.40%	3.2	Gage
132	Location of large diameter on taper		0.40%	3.2	Gage
133	Location of small diameter on taper		0.40%	3.2	Gage
134	Concentricity of taper with Datum A		0.40%	3.2	Gage
135	Overall length		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Check the source to verify that this is the current version before use.
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39:42

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Housing, Delay, Dummy		9298012
			NEXT HIGHER ASSEMBLY 9236005

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Knurl missing		0.40%	3.2	Visual
132	Location of small diameter on taper		0.40%	3.2	Gage
133	Location of large diameter on taper		0.40%	3.2	Gage
134	Circular runout of small diameter on taper		0.40%	3.2	Gage
135	Circular runout of large diameter on taper		0.40%	3.2	Gage
136	Depth of O-ring groove		0.40%	3.2	Gage
Minor					
201	O.D. of step below Knurl surface		0.65%	3.2	Gage
202	Diameter of surface before knurl		0.65%	3.2	Gage
203	Distance, large end surface to chamfer at large outside diameter.		0.65%	3.2	Gage
204	Width of O-ring groove		0.65%	3.2	Gage
205	Location of O-ring groove		0.65%	3.2	Gage
206	Corner radii of O-ring groove missing		0.65%	3.2	Visual
207	Surface finish of O-ring groove and taper		0.65%	3.2	Visual
208	Overall length		0.65%	3.2	Gage
209	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Check the source to verify that this is the current version before use. https://assist.dla.mil Downloaded: 2026-10-22T19:39:2

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Warhead, prior to loading	1 OF 1	9236005
			NEXT HIGHER ASSEMBLY
			-

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Foreign matter in charge cavity		0.40%	3.2	Visual
132	Unsolidified pool of paint in bottom of cavity		0.40%	3.2	Visual/Manual
133	Interior protective coating with bare spot or blister		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Warhead loaded assembly prior to assembly of fuze	1 OF 1	9236005
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Foreign matter in fuze well		0.40%	3.2	Visual
Minor					
201	Depth of fuze well		0.65%	3.2	Gage
202	Diameter of fuze well		0.65%	3.2	Gage
203	Concentricity of fuze well with datum A		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.00%	3.2	Visual

Check the source to verify that this is the current version before use. Downloaded: 2026-10-22T19:39Z https://assist.dla.mil

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Dummy Delay prior to Assembly to M.B.				9236005
					NEXT HIGHER ASSEMBLY
					-
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	O-ring missing		0.40%	3.2	Visual
132	Grease or lubricant not on O-ring		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship.		1.00%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39:02
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Motor body-Dummy delay assembly prior to loading of aft grain	1 OF 1	9236005
			NEXT HIGHER ASSEMBLY: -

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major B					
131	Ring seal missing or not pressed		0.40%	3.2	Visual
132	Nozzle insert missing or inverted		0.40%	3.2	Visual
133	Foreign material in cavity		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30:
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7.1	Motor body prior to loading of forward dummy propellant grain	1 OF 1	9236005
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Aft grain cushion missing prior to assembly of grain support bonding assembly.		0.40%	3.2	Visual
132	Grain Support bonding assembly missing.		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Check the source to verify that this is the current version before use. Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Motor body and Warhead prior to assembly	1 OF 1	9236005
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	O-ring missing from Warhead		0.40%	3.2	Visual
132	Warhead insulation missing		0.40%	3.2	Visual
133	Threads not free from grease or foreign material prior to application of adhesive		0.40%	3.2	Visual
134	Grain cushion missing or improperly bonded		0.40%	3.2	Visual
135	Silicone conpound missing or improperly applied to O-ring and Bearing surface		0.40%	3.2	Visual
Minor					
201	Primer or adhesive missing or improperly applied to motor body threads.		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Source: https://assistdia.mil -- Downloaded: 2026-10-22T19:39:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Inert Round Assembly		9236005
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE, INSPECTION METHOD
Critical					
1.	Obturator missing or damaged		100%	3.2	Visual
Major A					
101	Excessive gap between warhead and motor body		100%	3.2	Gage
102	Projectile fails to pass through ring gage		100%	3.2	Gage
Major B					
131	Rotating band nicked or damaged		0.40%	3.2	Visual
132	Fuze not fully seated		0.40%	3.2	Visual/Manual
Minor					
201	Painted surface with total exposed or damaged area in excess of 1/4 square inch.		0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable including incorrect color code		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.2	Visual

NOTES:

Source: http://assistdla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 ASSEMBLY TORQUE (See 3.4 and dwg. 9236005). This test shall be performed 100% on all projectiles. A thin line shall be drawn across the joint in question to aid in observing movement. If any projectile fails to comply with the torque requirement it shall be classed defective and removed from the lot.

Motor Body - Warhead (Relative movement upon reapplication of minimum assembly torque in assembly direction) Major A

4.4.3.2 DISASSEMBLY TORQUE (See 3.4.1)-This test shall be performed 100% on all projectiles. A thin line shall be drawn across the joint in question to aid in observing movement. A torque in accordance with 3.4.1 shall be applied in the disassembly direction and if any projectile fails to comply with the torque requirement it shall be classified defective and removed from the lot.

Warhead - Lifting Plug (Relative movement upon application of minimum disassembly torque in disassembly direction). Major A

4.4.3.3 X-ray examination for missing or improperly expanded shear ring seal. Major "A" Defect.-Each motor body assembly shall be subjected to a single x-ray examination in a plane that will clearly define the ring seal between the motor body and delay. The shear ring seal shall be fully expanded in the delay and motor body grooves and the groove in the motor body shall be completely filled with lead. There shall be no gap between the shear ring seal and the shoulder of the expansion cone. If any assembly shows evidence of a gap it shall be classified defective and removed from the lot.

4.4.4 Inspection Equipment.-The contractor shall design inspection equipment required to perform the examinations and tests prescribed in this section in accordance with 6.2.

4.4.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements stated herein may be used by DOD activities for design, procurement, manufacture, testing evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4.4.2 Supply and maintenance.-Supply and maintenance of the inspection equipment shall be in accordance with Specification [illegible].

4.4.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.5 Test Methods and procedures.

4.5.1 Determination for missing or improperly expanded shear ring seal.-The projectile shall be subject to X-ray examination (See 4.4.3.3). In all exposures the x-ray beam shall form a 90° angle with the longitudinal axis of the projectile. Examination shall be made to determine compliance with the applicable requirement.

4.5.2 Radiograph equipment, operations and procedures.-Radiographic equipment, operations and procedures shall be qualified in accordance with standards MIL-STD-453.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with the contract and MIL-STD-1169, Level C (Commercial Packaging).

6. NOTES

6.1 Ordering Data.-See MIL-A-48078 (PA)

6.2 Submission of inspection equipment designs for approval.-See MIL-A-48078 (PA). Submit equipment designs as required to Commanding Officer, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey, 07801.

6.3 It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.4 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination the characteristics may be measured or gaged to determine acceptability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.

6.5 Submission of certificates of conformance and ammunition data cards.-In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all certificates of conformance and ammunition data cards shall be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801. SARPA-QA-A-A.

Custodian:
ARMY - PA

Preparing Activity:
ARMY - PA

Project Number: 1320-A-276

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:39Z
Check the source to verify that this is the current version before use.